

VSEPR-SIM THEORETICAL DIVISION
TOP SECRET — Internal Development Document

XYZ File Format Family

Canonical Reference for `.xyz`, `.xyza`, `.xyzc`,
`.xyzf`, and `.xyzFull`

Grammar | Unit conventions | PBC & lattice mathematics |
Bond detection | Trajectory analysis | Provenance hashing |
Validation | Interoperability | Implementation mapping

VSEPR-SIM Development Reference

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§1 Overview

VSEPR-SIM uses a family of text-based coordinate files derived from the standard XYZ format. All variants are line-oriented, human-readable, and append-compatible. The family is ordered by information density:

Extension	Name	Per-atom data	Primary use
<code>.xyz</code>	Standard XYZ	species, position	Import/export baseline; external tool interoperability
<code>.xyza</code>	Extended XYZ	+ charge, velocity, force	Production trajectory output
<code>.xyzc</code>	Checkpoint XYZ	+ full header block	Simulation restart; checkpoint/restart files
<code>.xyzf</code>	Formation XYZ	species, position (multi-frame)	Formation trajectories; NEB images; defect snapshots
<code>.xyzFull</code>	Full-State XYZ	+ stress, KE, PE, image count	Rich replay; analysis pipeline carrier

Table 1: Format hierarchy.

All formats share the same per-atom coordinate block grammar. They differ in comment-line content, per-atom column set, and optional header blocks. A file in any format is a valid read target for any parser one level below it in the hierarchy; extra columns are discarded gracefully.

1.1 Unit Conventions

Quantity	Unit	Symbol	Notes
Position	Ångström	Å	
Velocity	Å/femtosecond	Å fs ⁻¹	
Force	eV/Å	eV Å ⁻¹	
Charge	elementary charge	<i>e</i>	
Mass	atomic mass units	amu	
Energy (total)	electronvolt	eV	per cell
Energy (per atom)	electronvolt	eV atom ⁻¹	in <code>.xyzFull</code>
Time	femtosecond	fs	
Stress	eV/Å ³	eV Å ⁻³	<code>.xyzFull</code> only
Temperature	kelvin	K	comment line
Pressure	gigapascal	GPa	header only

Table 2: Unit system (all formats).

Units are implicit — never written into the file. All readers and writers in the VSEPR-SIM codebase assume the above system without conversion. Callers receiving data from external tools must convert before passing frames to VSEPR-SIM APIs.

1.2 Coordinate System

All formats use a right-handed Cartesian coordinate system with the *z*-axis as the primary symmetry axis for layered and planar systems. The origin is not fixed: positions are absolute within the simulation box, which may be centred at the origin or offset depending on the

lattice generator used. For periodic systems the box is defined by three lattice vectors **a**, **b**, **c** encoded in the comment line (see §7).

1.3 Reserved Species Tokens

In addition to standard element symbols, the following tokens are valid in the species column of any XYZ variant:

Token	Meaning
alpha	Alpha particle (${}^4\text{He}^{2+}$ proxy)
bead	Coarse-grained polymer/bead particle
ghost	Non-interacting probe particle
vacancy	Explicit vacancy marker (no mass, no force)
dummy	Geometry placeholder; excluded from energy

Reserved tokens are case-sensitive. They resolve to zero mass and zero covalent radius unless overridden in `element_data.hpp`.

1.4 Connection to the Artifact Stack

The XYZ family integrates with the wider VSEPR-SIM output stack defined in WO-CHART-FIELD-B:

XYZ format	Downstream consumers
.xyza	Analysis pipeline: RDF, MSD, coordination number, field projection; also .xchart trajectory views
.xyzc	Simulation restart; checkpoint integrity hash
.xyzf	Formation event timeline in .dynx ; NEB chart projections in .xchart
.xyzFull	Rich replay in Qt desktop; full analysis pipeline; input to .xchart property surfaces

§2 Standard XYZ (.xyz)

The standard XYZ format is the baseline interoperability format. It is accepted by VESTA, Ovito, ASE, and most molecular visualisation tools without modification.

2.1 Grammar

```

1 xyz-file    ::= frame+
2 frame      ::= atom-count NEWLINE comment NEWLINE atom-line+
3 atom-count ::= INTEGER
4 comment    ::= STRING          (* free-form; parsed by VSEPR-SIM
   convention *)
5 atom-line  ::= SPECIES REAL REAL REAL NEWLINE
6 SPECIES    ::= element-symbol | reserved-token

```

Listing 1: Standard XYZ grammar (EBNF sketch).

Fields are whitespace-delimited. Tabs and multiple spaces are both accepted. Trailing whitespace is ignored. Blank lines between frames are not permitted; they cause a parse error. The parser is strict on frame boundaries but lenient on internal whitespace.

2.2 Comment Line Convention

VSEPR-SIM stores structured metadata in the comment line using a **key=value** syntax. Keys are lowercase and space-separated. Values containing spaces must be enclosed in double quotes.

```

1 step=1500 time_fs=75.000 energy_eV=-342.819 temperature_K=301.4
2   pbc="T T T" lattice="22.624 0.0 0.0 0.0 22.624 0.0 0.0 0.0
   22.624"
3   label=nacl_md hash=a3f82c91

```

Listing 2: Full comment line example.

Key	Type	Meaning	Formats
step	int	Simulation step index	all
time_fs	float	Simulation time (fs)	all
energy_eV	float	Total system energy (eV)	all
temperature_K	float	Instantaneous kinetic temperature (K)	all
pbc	string	"T T T" per-axis PBC flags	all
lattice	string	3×3 box matrix, row-major (Å)	all
label	string	Free-text run label	all
hash	string	Frame provenance hash (8 hex chars)	all
frame_type	string	Physical frame label (see §5)	.xyzf
Properties	string	ASE column declaration	.xyza
ke_eV	float	Total kinetic energy (eV)	.xyzFull
pe_eV	float	Total potential energy (eV)	.xyzFull
virial_eV	float	Scalar virial (eV)	.xyzFull

Table 3: Recognised comment-line keys.

Unknown keys are silently ignored, enabling forward compatibility.

2.3 Example: Water

```

1 3
2 step=0 time_fs=0.000 energy_eV=-14.203 label=H2O_relax
3 O 0.000000 0.000000 0.119748
4 H 0.000000 0.756950 -0.478993
5 H 0.000000 -0.756950 -0.478993

```

Listing 3: Minimal water molecule (.xyz).

2.4 Multi-Frame Trajectory

Multiple frames are concatenated without separators. A reader must consume exactly `atom_count` atom lines per frame before beginning the next frame header. The total number of frames is not known in advance; readers should stream rather than pre-allocate.

Frame count given file size F (bytes), average bytes per atom line $\bar{b}$, and comment/header overhead c :

$$N_{\text{frames}} \approx \frac{F}{c + N_{\text{atoms}} \bar{b}} \quad (1)$$

This estimate is approximate because $\bar{b}$ depends on coordinate precision and species name length. Use it only for pre-allocation hints.

2.5 Parsing Notes and Error Handling

- The atom count on line 1 is authoritative. Readers must not count atom lines manually.
- Species tokens are case-sensitive. `Na` and `na` are different tokens; only `Na` is valid.
- Coordinates beyond the fourth column are ignored in `.xyz` mode. They are meaningful only in `.xyza` and `.xyzFull`.
- An empty comment line is valid; all comment fields default to zero or empty string.
- A mismatch between the declared atom count and the number of atom lines before the next frame header is a **hard error**. The parser throws `XYZParseError` with the offending line number.
- Non-finite coordinate values (`nan`, `inf`) are treated as a hard error. The simulation has diverged and the file should not be silently passed to downstream tools.
- Lines beginning with `#` outside a checkpoint header block are treated as comments and skipped.

§3 Extended XYZ (`.xyza`)

The extended format appends per-atom dynamical quantities to each atom line. It is the primary output format for VSEPR-SIM production runs and the main input to the analysis pipeline.

3.1 Grammar

```

1 atom-line ::= SPECIES X Y Z Q VX VY VZ FX FY FZ NEWLINE
2           (* all numeric fields are REAL *)

```

Listing 4: Extended XYZ atom-line grammar.

Column order is fixed. A reader that encounters fewer than 11 data columns per atom line shall treat all missing trailing columns as zero. A reader encountering more than 11 columns shall ignore the excess unless it is operating in `.xyzFull` mode.

3.2 Property Columns

3.3 Properties Declaration Syntax

For interoperability with ASE extended-XYZ, VSEPR-SIM optionally writes a `Properties` key in the comment line:

```

1 Properties=species:S:1:pos:R:3:charge:R:1:vel:R:3:forces:R:3

```

Listing 5: Properties key (ASE compatibility).

The value is a colon-delimited list of `name:type:count` triples. Type codes: `S` = string, `R` = real, `I` = integer. This key is optional; VSEPR-SIM readers use the fixed column order regardless.

Col	Field	Symbol	Unit	Notes
1	Species	—	—	Element or reserved token
2	x	x	Å	Cartesian x position
3	y	y	Å	Cartesian y position
4	z	z	Å	Cartesian z position
5	Charge	q	e	Partial atomic charge
6	v_x	v_x	Å fs ⁻¹	Velocity x -component
7	v_y	v_y	Å fs ⁻¹	Velocity y -component
8	v_z	v_z	Å fs ⁻¹	Velocity z -component
9	F_x	F_x	eV Å ⁻¹	Force x -component
10	F_y	F_y	eV Å ⁻¹	Force y -component
11	F_z	F_z	eV Å ⁻¹	Force z -component

Table 4: Extended property columns (fixed order).

3.4 Example: NaCl Unit Cell

```

1 2
2 step=200 time_fs=10.000 energy_eV=-8.142 temperature_K=300.1
3   pbc="T T T" lattice="5.640 0.0 0.0 0.0 5.640 0.0 0.0 0.0 5.640"
4 Na  0.000000 0.000000 0.000000 +1.0 0.023 -0.011 0.008 0.312
   -0.197 0.143
5 Cl  2.820000 2.820000 2.820000 -1.0 -0.023 0.011 -0.008 -0.312
   0.197 -0.143

```

Listing 6: Two-ion NaCl unit at 300 K (.xyza).

3.5 Charge Semantics

The charge column carries the value from the active charge model:

Charge model	Source	Typical range	Continuity
formal	Integer valence assignment	$\{-4, \dots, +4\}$	discrete
neutral	Zero for all species	0	—
bader	Bader partitioning	$\sim \pm 2$	continuous
eam	Not applicable (metallic)	0	—

The active charge model is determined by the [material] registry field `default_charge_model` (see `VSIM_LANGUAGE.md`). Mixing charge models across frames of the same trajectory is a validation error (see §11).

§4 Checkpoint XYZ (.xyzc)

The checkpoint format extends .xyza with a structured header block carrying all state required to resume a simulation without re-reading the original script. A single .xyzc file is a complete restart package.

4.1 Header Block

The header block precedes the first frame. It begins with `## XYZC HEADER BEGIN` and ends with `## XYZC HEADER END`. All interior lines begin with `#` and use `key = value` syntax.


```

1 ## XYZC HEADER BEGIN
2 # format_version = 2
3 # project        = nacl_relax_v3
4 # vsim_version   = v5.0.0-beta.9
5 # seed_base      = 1001
6 # step           = 4000
7 # time_fs        = 200.000
8 # dt_fs          = 0.500
9 # run_mode       = md
10 # solver         = verlet
11 # forcefield     = ewald_formal
12 # temperature_K  = 300.0
13 # pressure_GPa   = 0.0
14 # periodic       = true
15 # pbc_flags      = T T T
16 # lattice_ang    = 22.624 0.000 0.000 0.000 22.624 0.000 0.000
17 # n_atoms        = 512
18 # charge_model   = formal
19 # label          = nacl_4x4x4_md_300K
20 # file_hash      = a3f82c91d48e7b02
21 ## XYZC HEADER END
22 512
23 step=4000 time_fs=200.000 energy_eV=-17641.320 temperature_K=299.8
24 Na 0.000000 0.000000 0.000000 +1.0 ...

```

Listing 7: Checkpoint header block (.xyzc).

4.2 Header Field Semantics

All fields are written by VSEPR-SIM. A reader encountering an absent field falls back to the current script’s values.

4.3 Restart Protocol

1. Parse the **## XYZC HEADER** block into a **CheckpointHeader** struct.
2. Validate **n_atoms** against the subsequent frame’s atom count. Mismatch is a hard error.
3. Validate the **file_hash** field against a freshly computed hash of the file’s first 10 KB (integrity check).
4. Override the active script’s **[run]** and **[simulation]** fields with the header values: step counter, timestep, solver, force-field, PBC flags, and lattice.
5. Load positions, velocities, forces, and charges directly into the integrator’s state arrays.
6. Resume the time loop from **step + 1**.

Note. Checkpoint restart does not re-run the presolve or lattice-builder phases. Geometry is taken entirely from the file. If the original lattice-builder would produce a different geometry (e.g. due to a seed change), the checkpoint takes priority.

Field	Type	Meaning
<code>format_version</code>	int	Header schema version (currently 2)
<code>project</code>	string	Script <code>[project].name</code> value
<code>vsim_version</code>	string	VSEPR-SIM build version string
<code>seed_base</code>	int	RNG seed used for the run
<code>step</code>	int	Last completed step index
<code>time_fs</code>	float	Simulation time at checkpoint (fs)
<code>dt_fs</code>	float	Integration timestep (fs)
<code>run_mode</code>	string	<code>relax</code> , <code>md</code> , <code>npt</code> , ...
<code>solver</code>	string	Active integrator tag
<code>forcefield</code>	string	Active force-field tag
<code>temperature_K</code>	float	Target temperature (K)
<code>pressure_GPa</code>	float	Target pressure (GPa; NPT only)
<code>periodic</code>	bool	PBC enabled at checkpoint
<code>pbs_flags</code>	string	"T T T" per-axis flags
<code>lattice_ang</code>	string	3×3 lattice matrix, row-major (Å)
<code>n_atoms</code>	int	Particle count (must match frame)
<code>charge_model</code>	string	<code>formal</code> , <code>neutral</code> , <code>bader</code>
<code>label</code>	string	Free-text run identifier
<code>file_hash</code>	string	SHA-256 prefix of the full file at write time

Table 5: Checkpoint header fields.

4.4 Checkpoint vs. Trajectory

A `.xyzc` file written every N steps doubles as a trajectory: the header block appears once at the start; subsequent frames are appended as plain `.xyza` blocks. To seek to frame k , a reader computes the line offset:

$$\text{offset}(k) = N_{\text{header}} + k \times (N_{\text{atoms}} + 2) \quad (2)$$

where N_{header} is the number of header lines (including the two sentinel lines), and the +2 accounts for the atom-count line and the comment line per frame. This is deterministic given `n_atoms`.

§5 Formation XYZ (`.xyzf`)

The formation format records the positional trajectory of a formation or reaction event as a sequence of standard coordinate frames. It omits velocities and forces to keep file size minimal.

5.1 Grammar

```

1 xyzf-file  ::= frame+
2 frame      ::= atom-count NEWLINE comment NEWLINE atom-line+
3 atom-line  ::= SPECIES X Y Z NEWLINE

```

Listing 8: Formation XYZ grammar.

The grammar is syntactically identical to `.xyz`. The semantic distinction is that the comment line always carries a `frame_type` key indicating the physical role of the frame.

5.2 Comment Line Convention

```

1 frame=0 frame_type=pre_formation step=0 energy_eV=-342.100
2 frame=1 frame_type=transition step=45 energy_eV=-339.817
3 frame=2 frame_type=post_formation step=80 energy_eV=-346.203

```

Listing 9: Formation XYZ comment lines over a reaction event.

frame_type token	Meaning
pre_formation	Initial, unformed state
transition	Intermediate or saddle-point frame
post_formation	Final, formed state
defect_snap	Defect snapshot during a longer trajectory
neb_image	Nudged-elastic-band intermediate image
arbitrary	No specific physical label assigned

5.3 Reaction Coordinate

For NEB pathways and reaction trajectories, VSEPR-SIM writes a `rxcoord` key in the comment line encoding the reaction coordinate $\xi \in [0, 1]$:

$$\xi_k = \frac{\sum_{j=1}^k \|\mathbf{r}^{(j)} - \mathbf{r}^{(j-1)}\|}{\sum_{j=1}^M \|\mathbf{r}^{(j)} - \mathbf{r}^{(j-1)}\|} \quad (3)$$

where $\mathbf{r}^{(k)}$ is the concatenated position vector of all atoms at frame k , and $M + 1$ is the total number of frames. This maps the discrete image sequence onto a normalised path length.

5.4 Frame Independence and Variable Atom Counts

Each frame is fully self-contained. The atom count may differ between frames (e.g. if atoms are inserted or removed during a formation event). Readers must re-parse the atom count at the start of each frame and must not assume count consistency. If frame-to-frame count changes are detected, a `variable_count` flag is emitted in the metadata.

5.5 Typical Uses

Use case	Description
Formation event recording	Capture atomic positions during bond formation
Defect migration path	Snapshot sequence along a migration trajectory
NEB pathway	Nudged-elastic-band intermediate images
Visualisation export	Compact trajectory for external viewers
Batch comparison	Multiple independent frames in one file
Reaction pathway archive	Store full reaction path with energy profile

Table 6: Common `.xyzf` applications.

§6 Full-State XYZ (.xyzFull)

The .xyzFull format is the richest member of the family. It extends .xyza with per-atom stress contributions, kinetic and potential energy decompositions, and PBC image counters. Its primary roles are rich trajectory replay in the Qt desktop and serving as the input carrier for the full analysis pipeline (RDF, MSD, field projection, scale sampling).

6.1 Grammar

```

1 atom-line ::= SPECIES  X Y Z  Q  VX VY VZ  FX FY FZ
2                KE  PE  SXX SYX SZZ SXY SXZ SYZ
3                IX IY IZ  NEWLINE
4 (* 22 fields per line: 1 string + 21 reals *)

```

Listing 10: .xyzFull atom-line grammar.

Col(s)	Field	Symbol	Unit	Notes
1	Species	—	—	
2–4	Position	x, y, z	Å	
5	Charge	q	e	
6–8	Velocity	$\mathbf{v}$	Å fs ⁻¹	
9–11	Force	$\mathbf{F}$	eV Å ⁻¹	
12	KE per atom	ε_k	eV	$\frac{1}{2}mv^2$
13	PE per atom	ε_p	eV	force-field derived
14–19	Stress tensor	$\sigma_{\alpha\beta}$	eV Å ⁻³	Voigt order: xx, yy, zz, xy, xz, yz
20–22	Image count	n_x, n_y, n_z	—	PBC image displacement

Table 7: .xyzFull property columns (fixed order).

6.2 Stress Tensor Convention

The per-atom stress tensor σ_i is written in Voigt notation: $(\sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{xy}, \sigma_{xz}, \sigma_{yz})$.

The virial pressure is recovered from the per-atom contributions:

$$P = -\frac{1}{3V} \sum_{i=1}^N (\sigma_{xx,i} + \sigma_{yy,i} + \sigma_{zz,i}) \quad (4)$$

where $V = |\det \mathbf{H}|$ is the cell volume. This is written to the comment line as **virial_eV** for convenience.

6.3 Image Counters

For PBC trajectories, each atom accumulates an image displacement counter $(n_x, n_y, n_z) \in \mathbb{Z}^3$ tracking how many times it has crossed each periodic boundary. The *unwrapped* position is:

$$\mathbf{r}_i^{\text{unwrap}} = \mathbf{r}_i^{\text{wrap}} + \mathbf{H}^\top \mathbf{n}_i \quad (5)$$

Unwrapped positions are required for correct MSD computation (see §9.2).

6.4 Comment Line (Full-State)

The `.xyzFull` comment line carries all standard keys plus:

```

1 step=1000 time_fs=50.000 energy_eV=-17641.320 temperature_K=300.2
2 ke_eV=248.411 pe_eV=-17889.731 virial_eV=-52341.8
3 pbc="T T T" lattice="22.624 0.0 0.0 0.0 22.624 0.0 0.0 0.0
4 22.624"
   hash=b7d3a941

```

Listing 11: `.xyzFull` comment line.

Energy decomposition identity:

$$E_{\text{total}} = E_k + E_p, \quad E_k = \sum_{i=1}^N \varepsilon_{k,i}, \quad E_p = \sum_{i=1}^N \varepsilon_{p,i} \quad (6)$$

§7 Bounding Box and Periodic Boundary Conditions

All format variants support the same PBC encoding in the comment line. The encoding is activated when `pbc = "T T T"` (or any T/F subset) appears alongside a `lattice` key.

7.1 Comment-Line Encoding

```

1 pbc="T T T" lattice="22.624 0.0 0.0 0.0 22.624 0.0 0.0 0.0 22.624"

```

Listing 12: Orthorhombic box.

```

1 pbc="T T T" lattice="11.312 0.0 0.0 5.656 9.797 0.0 0.0 0.0
  22.624"

```

Listing 13: Triclinic box (monoclinic example).

```

1 pbc="T T F" lattice="22.624 0.0 0.0 0.0 22.624 0.0 0.0 0.0 60.000"

```

Listing 14: Slab geometry (periodic in x,y only).

7.2 Lattice Matrix

The nine values of the `lattice` key form the matrix $\mathbf{H}$ in row-major order:

$$\mathbf{H} = \begin{bmatrix} \mathbf{a}^\top \\ \mathbf{b}^\top \\ \mathbf{c}^\top \end{bmatrix} \in \mathbb{R}^{3 \times 3} \quad (7)$$

An atom at fractional coordinates $\mathbf{s} \in [0, 1]^3$ has Cartesian position $\mathbf{r} = \mathbf{H}^\top \mathbf{s}$. The cell volume is $V = |\det \mathbf{H}|$. For orthorhombic boxes, $\mathbf{H}$ is diagonal: $L_x = a_{11}$, $L_y = a_{22}$, $L_z = a_{33}$.

7.3 Reciprocal Lattice

$$\mathbf{H}^* = 2\pi(\mathbf{H}^\top)^{-1} \quad (8)$$

Rows of $\mathbf{H}^*$ are the reciprocal lattice vectors $\mathbf{a}^*$, $\mathbf{b}^*$, $\mathbf{c}^*$ satisfying $\mathbf{a} \cdot \mathbf{a}^* = 2\pi$, $\mathbf{a} \cdot \mathbf{b}^* = 0$, etc. Used internally by the Ewald summation module; not written to XYZ files.

7.4 Fractional Coordinate Conversion

Cartesian to fractional:

$$\mathbf{s} = (\mathbf{H}^\top)^{-1} \mathbf{r} \quad (9)$$

Wrapping a position into the primary cell:

$$\mathbf{s}_{\text{wrap}} = \mathbf{s} - \lfloor \mathbf{s} \rfloor \quad (10)$$

7.5 Minimum Image Convention

$$\Delta \mathbf{r}_{ij}^{\text{MIC}} = \Delta \mathbf{r}_{ij} - \mathbf{H}^\top \left\lfloor \mathbf{H}^{-\top} \Delta \mathbf{r}_{ij} + \frac{1}{2} \right\rfloor \quad (11)$$

For orthorhombic boxes:

$$\Delta r_{\alpha,ij}^{\text{MIC}} = \Delta r_{\alpha,ij} - L_\alpha \text{round}\left(\frac{\Delta r_{\alpha,ij}}{L_\alpha}\right) \quad (12)$$

pbcs = "T T F" applies PBC in x and y only (slab geometry).

§8 Bond Auto-Detection and Neighbor Lists

8.1 Bond Detection Criterion

$$\text{bonded}(i, j) \iff d_{ij}^{\text{MIC}} \leq \lambda(r_{\text{cov}}^{(i)} + r_{\text{cov}}^{(j)}) \quad (13)$$

Default $\lambda = 1.15$; overridable via `[simulation].bond_detect_scale`.

8.2 Neighbor List Construction

For $N > 500$ atoms, bond detection uses a cell-list neighbor structure. Cut-off radius:

$$r_{\text{cut}} = \lambda \max_k (2r_{\text{cov}}^{(k)}) + 0.3 \text{ \AA} \quad (14)$$

Cells per axis:

$$n_\alpha = \left\lceil \frac{L_\alpha}{r_{\text{cut}}} \right\rceil \geq 1 \quad (15)$$

Only atoms in the same cell or 26 neighboring cells need distance evaluation, reducing complexity from $O(N^2)$ to $O(N)$.

8.3 Coordination Number

$$Z_i = |\{j \neq i : \text{bonded}(i, j)\}| \quad (16)$$

Mean coordination for species s :

$$\bar{Z}_s = \frac{1}{N_s} \sum_{i:\text{species}(i)=s} Z_i \quad (17)$$

Expected values: NaCl $\bar{Z} = 6$; diamond Si $\bar{Z} = 4$; graphene C $\bar{Z} = 3$; BCC Fe $\bar{Z} = 8$.

8.4 Design Rationale

- No pre-defined connectivity table is required.
- λ is a single global knob. Ionic systems may need $\lambda \approx 1.3$; covalent systems use $\lambda = 1.15$.
- For metallic systems, bond detection is suppressed (`bond_detect_scale = 0.0`).
- For bead/polymer systems, topology is injected by the bead-builder.
- Covalent radius table in `include/vsim/element_data.hpp` covers $Z = 1$ –118.

§9 Trajectory Analysis

9.1 Radial Distribution Function

$$g(r) = \frac{V}{N^2} \left\langle \sum_i \sum_{j \neq i} \delta(r - d_{ij}^{\text{MIC}}) \right\rangle \quad (18)$$

Discretised into bins of width Δr :

$$g(r_k) = \frac{V}{N^2} \cdot \frac{\langle n_k \rangle}{4\pi r_k^2 \Delta r} \quad (19)$$

Truncated at $r_{\text{max}} = L_{\text{min}}/2$ to avoid double-counting. Species-resolved partial RDF:

$$g_{AB}(r) = \frac{V}{N_A N_B} \left\langle \sum_{i \in A} \sum_{j \in B} \delta(r - d_{ij}^{\text{MIC}}) \right\rangle \quad (20)$$

9.2 Mean Squared Displacement

Requires unwrapped positions $\mathbf{r}_i^{\text{unwrap}}(t)$ from image counters in `.xyzFull`:

$$\text{MSD}(\tau) = \frac{1}{N} \left\langle \sum_{i=1}^N \|\mathbf{r}_i^{\text{unwrap}}(t_0 + \tau) - \mathbf{r}_i^{\text{unwrap}}(t_0)\|^2 \right\rangle_{t_0} \quad (21)$$

Solid: $\text{MSD}(\tau) \rightarrow C$ (bounded). Diffusing liquid: $\text{MSD}(\tau) \sim 6D\tau$ where D is the self-diffusion coefficient.

9.3 Velocity Autocorrelation and VDOS

Normalised VACF:

$$C_v(\tau) = \frac{\langle \sum_i \mathbf{v}_i(t_0 + \tau) \cdot \mathbf{v}_i(t_0) \rangle_{t_0}}{\langle \sum_i \|\mathbf{v}_i(t_0)\|^2 \rangle_{t_0}} \quad (22)$$

Vibrational density of states (VDOS):

$$g(\nu) = \int_{-\infty}^{\infty} C_v(\tau) e^{-2\pi i \nu \tau} d\tau \quad (23)$$

Requires velocities; `.xyz` and `.xyzf` cannot be used.

§10 Provenance Hashing

10.1 Frame Hash

$$H_{\text{frame}} = \text{SHA256}(\text{step} \parallel \text{n_atoms} \parallel \text{species} \parallel \text{positions})[0 : 8] \quad (24)$$

Position values are rounded to 6 decimal places before hashing.

10.2 File Hash

$$H_{\text{file}} = \text{SHA256}(\text{file}[0 : 10240]) \quad (25)$$

Recomputed on restart; mismatch is a hard error.

10.3 Hash Chain

$$H_k = \text{SHA256}(H_{k-1} \parallel H_{\text{frame},k})[0 : 8] \quad (26)$$

Opt-in via `[kernel].symbolic_trace = true`; stored in `write_pipeline_audit_jsonl`.

§11 Per-Format Validation

11.1 Universal Checks

1. Atom count on line 1 must equal the number of atom lines per frame.
2. All coordinate values must be finite (no NaN/Inf).
3. Species column must not be empty.
4. `step` ≥ 0 ; `time_fs` ≥ 0 .

11.2 .xyza Checks

1. Charge model must not change between frames.
2. Velocity and force values must be finite.
3. $\|\mathbf{v}_i\| < 100 \text{ \AA fs}^{-1}$ (warning above this).
4. $\|\mathbf{F}_i\| < 10^4 \text{ eV \AA}^{-1}$ (warning above this).

11.3 .xyzc Checks

1. Matching BEGIN/END sentinels required.
2. `n_atoms` must equal the first frame's atom count.
3. `file_hash` must match recomputed value on restart.
4. `format_version` must be $\leq$ reader's maximum.

11.4 .xyzFull Checks

1. $|E_{\text{total}} - (E_k + E_p)| < 10^{-6} \text{ eV}$.
2. $\varepsilon_{k,i} \geq 0$ for all atoms.
3. Stress Voigt components self-consistent.
4. $|\Delta n_\alpha| \leq 1$ per atom per axis between consecutive frames.

§12 Interoperability

12.1 ASE

ASE `extxyz` is compatible with `.xyza` when the `Properties` key is present. Column mapping: `pos` $\leftrightarrow x, y, z$; `charge` $\leftrightarrow q$; `vel` $\leftrightarrow v_x, v_y, v_z$; `forces` $\leftrightarrow F_x, F_y, F_z$. Stress and image counters are not mapped.

12.2 VESTA

VESTA reads standard `.xyz`. Export with `XYZFormat::Xyz` and `suppress_props = true`.

12.3 Ovito

Ovito reads `.xyz` and `.xyza` via its extended-XYZ parser. It accepts both uppercase `Lattice` and lowercase `lattice`; VSEPR-SIM writes lowercase.

§13 Export Profile Connection

[export] flag	File written	Format
<code>write_xyz = true</code>	<code><prefix>.xyz</code>	<code>.xyz</code>
<code>write_xyzf = true</code>	<code><prefix>.xyzf</code>	<code>.xyzf</code>
<code>write_xyzFull = true</code>	<code><prefix>.xyzFull</code>	<code>.xyzFull</code>
<code>write_checkpoint = true</code>	<code><prefix>_step{N}.xyzc</code>	<code>.xyzc</code>

Export profile expansion: `minimal` $\rightarrow$ `.xyz` only; `standard` $\rightarrow$ `.xyz`, `.xyzf`; `research_report` $\rightarrow$ adds `.xyzFull`; `publication` $\rightarrow$ all + `.xyzc` every 1000 steps.

§14 Implementation Mapping

Source file	Responsibility	Format(s)
<code>xyz_io.hpp</code> / <code>.cpp</code>	Core reader/writer	<code>.xyz</code> , <code>.xyza</code>
<code>xyz_full.hpp</code> / <code>.cpp</code>	Full-state reader/writer	<code>.xyzFull</code>
<code>xyz_checkpoint.hpp</code> / <code>.cpp</code>	Checkpoint I/O	<code>.xyzc</code>
<code>xyz_formation.hpp</code> / <code>.cpp</code>	Formation trajectory I/O	<code>.xyzf</code>
<code>element_data.hpp</code>	Covalent radius & mass table	all
<code>pbs_utils.hpp</code>	MIC, wrap/unwrap, cell volume	all
<code>cell_list.hpp</code>	Neighbor-list construction	all
<code>bond_detect.hpp</code>	Bond auto-detection	all
<code>traj_analysis.hpp</code> / <code>.cpp</code>	RDF, MSD, VACF, VDOS	<code>.xyza</code> , <code>.xyzFull</code>
<code>provenance.hpp</code>	Frame and file hash utilities	all
<code>vsim_export.hpp</code> / <code>.cpp</code>	Export dispatch layer	all

Table 8: Source files implementing the XYZ format family.

14.1 Core Frame Struct

```

1 namespace vsim {
2
3 struct XYZFrame {
4     int n_atoms;
5     XYZFormat format;
6     CommentFields comment;
7     std::vector<std::string> species;
8     std::vector<Vec3> positions; // Ang
9     std::vector<double> charges; // e
10    std::vector<Vec3> velocities; // Ang/fs (.xyza+)
11    std::vector<Vec3> forces; // eV/Ang (.xyza+)
12    // .xyzFull only:
13    std::vector<double> ke_per_atom; // eV
14    std::vector<double> pe_per_atom; // eV
15    std::vector<Stress3> stress; // eV/Ang3, Voigt
16    std::vector<IVec3> image_counts; // PBC image
17    displacement
18 };
19 } // namespace vsim

```

Listing 15: Core frame struct (xyz_io.hpp).

14.2 Reader API

```

1 namespace vsim {
2
3 bool read_frame(std::istream& in, XYZFrame& out,
4               XYZFormat fmt = XYZFormat::Auto);
5
6 std::vector<XYZFrame> read_all(
7     const std::filesystem::path& p,
8     XYZFormat fmt = XYZFormat::Auto);
9
10 // Seek to frame k in a .xyzc file (O(1) line offset).
11 bool seek_frame(std::istream& in, int k,
12               int n_atoms, int n_header_lines);
13
14 } // namespace vsim

```

Listing 16: XYZ reader API.

14.3 Writer API

```

1 namespace vsim {
2
3 struct WriteOptions {
4     int precision_pos = 6;
5     int precision_other = 4;
6     bool suppress_props = false;
7     bool write_hash = true;
8     bool chain_hash = false;
9 };
10
11 void write_frame(std::ostream& out, const XYZFrame& frame,
12               XYZFormat fmt, const WriteOptions& opts = {});

```

```

13
14 void append_frame(const std::filesystem::path& p,
15                  const XyzFrame& frame,
16                  XyzFormat fmt, const WriteOptions& opts = {});
17
18 } // namespace vsim

```

Listing 17: XYZ writer API.

14.4 Analysis API

```

1 namespace vsim {
2
3 struct RdfResult {
4     std::vector<double> r;
5     std::vector<double> g_r;
6     double dr;
7 };
8
9 struct MsdResult {
10     std::vector<double> tau;
11     std::vector<double> msd;
12     std::string regime; // solid_bounded / diffusive / mixed
13     double D; // diffusion coeff (Ang^2/fs)
14 };
15
16 RdfResult compute_rdf(const std::vector<XyzFrame>& traj,
17                      double dr = 0.05, double r_max = 0.0);
18
19 MsdResult compute_msd(const std::vector<XyzFrame>& traj);
20
21 } // namespace vsim

```

Listing 18: Trajectory analysis API (traj_analysis.hpp).

§15 Typical Energy and Observable Ranges

System	N	E_{total} (eV)	T (K)	d_{NN} (Å)	$\bar{Z}$
H ₂ O (1 molecule)	3	−14 to −13	0–600	0.96	O:2
NaCl rocksalt (4×4×4)	512	−17700 to −17400	0–1200	2.70–2.85	6
Si diamond (3×3×3)	216	−900 to −860	0–1600	2.33–2.36	4
Graphene (96 C)	96	−550 to −530	0–2000	1.40–1.43	3
Fe BCC (4×4×4)	128	−820 to −800	0–1800	2.45–2.50	8
UO ₂ fluorite (2×2×2)	96	−1900 to −1800	0–3000	2.34	U:8
Al FCC (4×4×4)	256	−860 to −830	0–900	2.85–2.87	12
TiO ₂ rutile (2×2×4)	192	−2800 to −2700	0–2000	1.94	Ti:6

Energy. Per simulation cell, not per atom. Per-atom values in `.xyzFull` and `write_analysis_json`.

Temperature. Instantaneous kinetic temperature:

$$T = \frac{2E_k}{3Nk_B} \quad (27)$$

Fluctuates substantially for small N . Block-averaged temperature in `write_metrics_tsv`.
MSD plateaus (solid, 300 K): NaCl 0.05–0.15 Å²; Si 0.01–0.05 Å²; Fe 0.02–0.08 Å².
Values substantially above these suggest melting or a runaway trajectory.